Data Augmentation On Plant Leaf Disease Image Dataset Using Image Manipulation and Deep Learning Techniques

Arun Pandian J
Dept. of Computer Science and Engineering,
M.A.M. College of Engg. & Tech.,
Tiruchirappalli, India
arunpandian@mamcet.com

G Geetharamani
Dept. of Mathematics,
University College of Engineering,
Anna University - BIT Campus,
Tiruchirappalli, India
geeramdgl@rediffmail.com

Annette B
Dept. of Management Studies,
M.A.M. College of Engg. & Tech.,
Tiruchirappalli, India
ganncliff@gmail.com

***Abstract*—A large volume of data will increase the performance of machine learning algorithms and avoid overfitting problems. Collecting a large amount of training data in the agricultural field for designing plant leaf disease detection and diagnosis model is a highly challenging task which takes more time and resources. Data augmentation increases the diversity of training data for machine learning algorithms without collecting new data. In this article, augmented plant leaf disease datasets was developed using basic image manipulation and deep learning based image augmentation techniques such as Image flipping, cropping, rotation, color transformation, PCA color augmentation, noise injection, Generative Adversarial Networks (GANs) and Neural Style Transfer (NST) techniques. Performance of the data augmentation techniques was studied using state-of-the-art transfer learning techniques, for instance, VGG16, ResNet, and InceptionV3. An extensive simulation shows that the augmented dataset using GAN and NST techniques achieves better accuracy than the original dataset using a basic image manipulation based augmented dataset. Furthermore, a combination of Deep learning, Color, and Position augmentation dataset gives the maximum classification performance than all other datasets.**

Keywords- Data augmentation, Deep Learning, Generative Adversarial Networks, Image manipulation, Neural Style Transfer.

I. INTRODUCTION

Identification Plant leaf diseases are most important to increase the farmer's benefits in terms of crop yield and growth. Continuous monitoring of diseases in plant leaf is more difficult and expensive to implement in real-time [1]. In recent years, several artificial intelligence methods have proposed different techniques to detect and diagnose plant leaf diseases using image processing techniques [2].

Typically, Deep Convolutional Neural Network (DCNN) plays an essential role in agriculture and medical image classification and detection [3, 9]. The DCNN needs a large volume of training data and a substantial computational resource. Advances in Graphical Processing Units (GPUs) can solve the high-performance computation requirement issues.

Data augmentation techniques can increase the quantity of data in a dataset without collecting new data. The data augmentation techniques create new data from the original training data using some basic manipulation and advanced image transformation techniques [28]. The most common image manipulation techniques are Image flipping, cropping, rotation, color transformation, and noise injection techniques. These techniques create augmented images using any one or combination of multiple image formatting methods [26].

Generative Adversarial Networks (GANs) and Neural Style Transfer (NST) techniques are advanced image transformation techniques and can create new augmented images using deep neural networks [18, 20]. Generative Adversarial Networks are the class of generative models, which create new unique data from the existing data in the dataset. Generator and discriminator are the two major neural network parts in the GAN [21]. In the training process, the Generator produces new data and tries to convince the Discriminator, and the Discriminator tries to compare the original and generated data. The training process will not be complete until the classification accuracy of the Discriminator gets very less.

Deep Convolutional Generative Adversarial Networks (DCGANs) is a class of unsupervised learning algorithms and uses convolutional neural networks (CNNs) to build the generator and discriminator [4]. DCGAN was used in several applications in the computer vision field to create unique images. The performance of the DCGAN is better than the traditional GAN network.

Moreover, Wasserstein GANs (WGANs) is another well-known GAN based learning approach [5]. The WGAN is a modified version of DCGAN. It uses a linear activation function instead of sigmoid in the output layer and Wasserstein loss function to calculate the loss values of the training process.

On the other hand, Neural Style Transfer (NST) is an advanced image transformation technique. It uses deep neural networks to perform the image transformation [6]. The NST uses two input images to generate output images, named as a content image and style image. It takes the input information from the content images and uses the design patterns of style images to create an output image. Mainly, it transforms the style into the content image to generate a new artistic output image [23, 24]. The NSTs are mostly used to create artificial paintings and designs.

The remaining part of this article is organizing as follows; Section 2 presents a detailed literature survey of deep learning based data augmentation techniques. Section 3 describes the original plant leaf disease dataset for plant leaf disease detection. Section 4 discusses the use of basic image manipulation techniques to create an augmented

978-1-7281-4392-7/19/$31.00 ©2019 IEEE

dataset. Section 5 discusses the implementation of Deep learning based data augmentation approaches. Section 6 provides the performance comparison of the developed datasets. As a final point, section 6 presents the conclusion and future works.

II. LITERATURE SURVEY

The most crucial objective of this survey is to illustrate the various deep learning techniques that are used to generate augmented images and the study of the significance of the augmentation process.

In [10], the authors proposed a liver lesion classification using GAN data augmentation. The augmented dataset based classification algorithm achieved the sensitivity and specificity of 78.6% and 88.4%, respectively. The classification algorithm using the GAN data augmentation attained a sensitivity of 85.7% and a specificity of 92.4%. These results are more excellent than the original dataset.

Similarly, the authors in [27] have proposed a liver lesion classification model using GAN-based data augmentation to achieve improved performance. In [29], the author proposed a DCNN model for liver lesion classification problem using DGAN based data augmentation technique. The generated images are evaluated using a visual Turing test [11]. Likewise, the authors in [30] created DCGAN and WGAN based augmented images for brain tumor MRI datasets. The authors evaluate the quality of the generated images using several performance metrics.

The authors in [12] developed a new GAN architecture, which improved the output image quality of the image-to-image translation process. The architecture was named CycleGAN, and it uses the Cycle-Consistency loss function to stabilize the training process of the networks.

Another architecture proposed by the authors in [7] named as Progressively Growing GANs, uses progressive resolution complexity to train the networks. This approach train the model with progressive resolution. For instance, the resolution range starts from the input size of 4 × 4 to 8 × 8, and so on until output resolution of 1024 × 1024 is achieved.

In [16], the authors compared the performance of several loss functions of generator and discriminator. The authors conclude that most of the loss functions give a similar performance with optimized hyperparameters.

In [25], the authors used GANs to create realistic data from their simulated data. CycleGAN based augmented dataset was designed to avoid class imbalance issues by the authors in [14]. It created an augmented facial expression database to recognize seven different facial emotions. Similarly, the performance of the CycleGANs was tested by the authors in [13] for the facial expression recognition task.

Another GAN architecture proposed by the authors in [15] is Conditional GANs for image augmentation. The Conditional GANs are using conditional vector in the generator and the discriminator to solve mode collapse issues. The authors in [17] review the applications and limitations of GANs in medical imaging applications.

In [19], the authors describe the challenges and drawbacks of training GANs for the data augmentation process. In [6], the authors propose an NST technique to transfer the image style to the input image using neural networks.

The authors in [22] have developed an NST algorithm to generate real-world images. They use the instance normalization function for the training process instead of batch normalization, which achieves better performance than standard NST technique. Authors in [8] developed the fastest NST algorithm to create augmented images. The drawback of the NST algorithm was limited style transfer.

III. ORIGINAL PLANT LEAF DISEASE DATASET

The original plant leaf disease dataset was download from the Mendeley open data repository [34]. The dataset contains 38 different healthy and diseased plant leaf classes of 54305 color images.

There are 13 different plant leaves used in the dataset. The number of images in every single class is not in a uniform range, and it shows that the data set is imbalanced. For instance, the orange Huanglongbing class contains 5507 images, and the Healthy potato class consists of only 152 images. The imbalanced dataset may affect the performance of the classification model.

Figure 1 shows the sample images from the original dataset.

Fig 1. Sample data from the original dataset

The data augmentation techniques are introduced to handle the imbalanced dataset. The basic image manipulation based augmentation techniques to solve the data imbalance issues discussed in the next section.

Figure 2 shows the imbalanced structure of the original dataset.

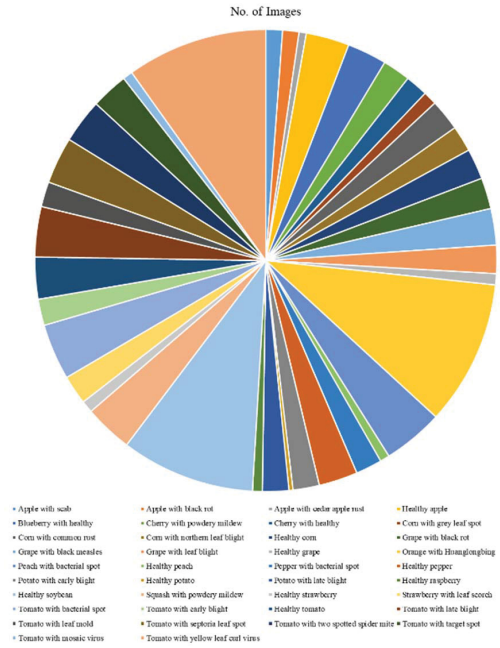

Fig 2. Structure of the original dataset

IV. DATA AUGMENTATION USING BASIC MANIPULATION TECHNIQUES

The most common image manipulation techniques are Image flipping, cropping, rotation, color transformation, principal component analysis (PCA), and noise injection. The Basic image manipulation techniques creates 32073 augmented images and it balanced each class size with 2000 images in the plant leaf disease dataset. Figure 3 shows sample augmented images using basic image manipulation techniques.

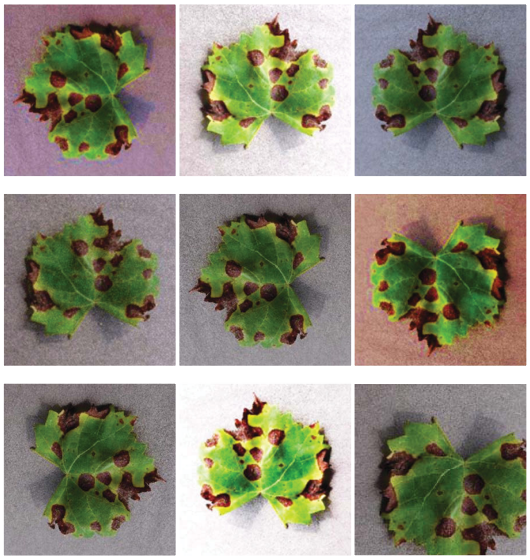

Fig 3. Sample augmented data using basic manipulation

The size of the augmented dataset was 76000 images. The augmented dataset was shared in the kaggle open data repository [32].

V. DATA AUGMENTATION USING DEEP LEARNING TECHNIQUES

Deep Convolutional Generative Adversarial Networks, Wasserstein GAN, and Neural Style Transfer are the most common Deep learning based data augmentation techniques. The standard DCGAN and WGAN were trained using NVIDIA DGX-1 V100 deep learning server with epochs of 1000 and batch size of 64.

The DCGAN and WGAN were generated 32073 augmented images to enhance the size of the original dataset. It balanced the dataset by the individual class size of 2000 images. Figure 4 illustrates the DCGAN and WGAN based sample augmented images. The augmented dataset was shared in the kaggle open data repository [33].

Fig 4. Sample augmented data using DCGAN and WGAN

The Neural Style Transfer (NST) augmentation technique creates 6038 images. The convolutional neural network was used to design the Neural Style Transfer model. The model was trained using the training epochs of 3000 and a minibatch size of 64.

The augmentated data equalized the individual class size as 1000 images. The augmented dataset size was 38,000 images. The augmented dataset was shared in the kaggle open data repository [35].

Figure 5 shows the random samples from the augmented dataset using the Neural Style Transfer technique.

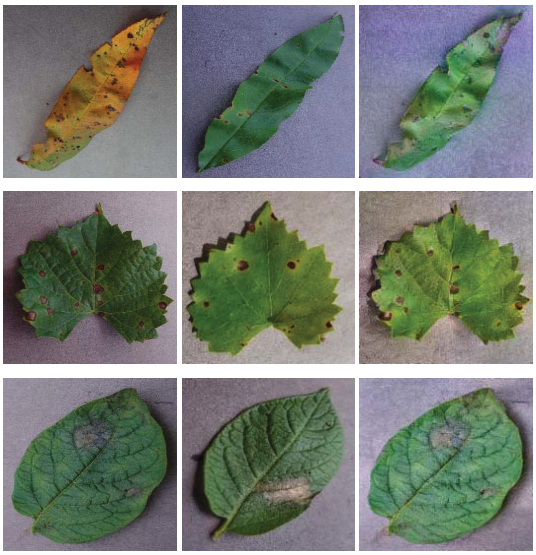

Fig 5. Sample augmented data using NST

VI. RESULTS AND DISCUSSIONS

At first, the original dataset was used to train the models for plant leaf disease identification with different state-of-the-art transfer learning techniques such as VGG16, ResNet, and InceptionV3.

The dataset divided into training, validation, and testing set at 90%, 5%, and 5%, respectively. Figure 6 shows the classification accuracy of the models using original dataset.

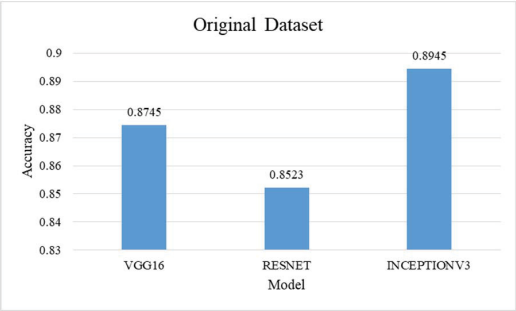

Fig 6. Performance of different transfer learning techniques using the original dataset

Moreover, the basic image manipulation technique based dataset was used to train the pre-trained models with similar hyperparameters value. The testing result shows that the augmented dataset gives a better classification performance than the original dataset.

Figure 7 shows the performance of the different models using the basic image manipulation based augmented dataset.

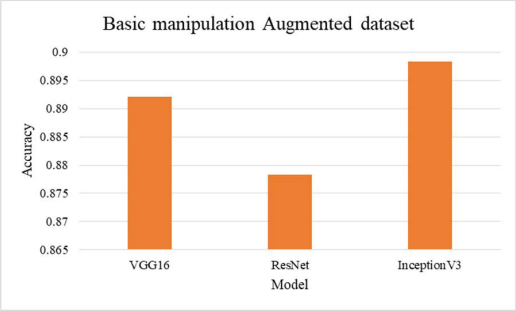

Fig 7. Performance of different transfer learning techniques using the basic image manipulation based augmented dataset

Furthermore, the DCGAN and WGAN augmented dataset based classification models are trained using the value of the same hyperparameter. After the training process, the models are tested using unseen testing images. The testing results show that the performance of the classification models using the DCGAN and WGAN augmented dataset gives a superior performance than the previous datasets. Figure 8 shows the performance of the DCGAN and WGAN augmented dataset.

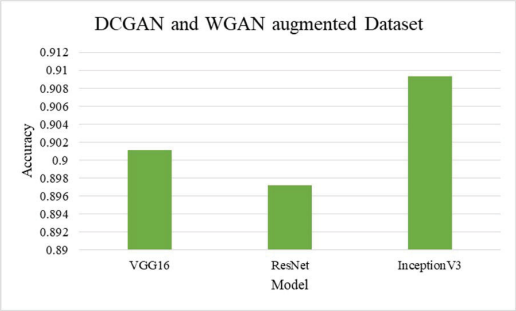

Fig 8. Performance of different transfer learning techniques using the DCGAN and WGAN augmented dataset

Likewise, the pre-trained models are trained using the NST augmented dataset. The testing result shows that the NST augmented dataset gives better accuracy than the basic manipulation dataset and original dataset. However, the performance is quite less than the DCGAN and WGAN augmented dataset. Figure 9 shows the testing results of the dataset.

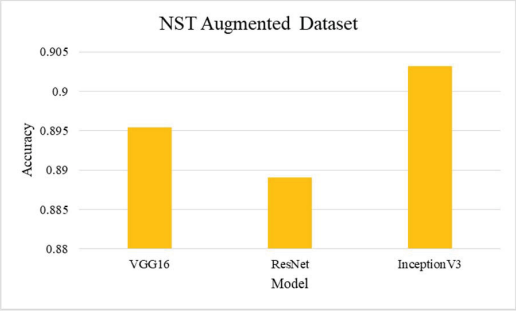

Fig 9. Performance of different transfer learning techniques using the NST augmented dataset

Finally, the combined dataset was created by using all the augmentation techniques such as basic manipulation,

DCGAN and WGAN, and NST. The size of the dataset was 228000. The classification accuracy of the pre-trained models using the dataset was much higher than other datasets. Figure 10 shows the classification accuracy of the pre-trained models using the dataset.

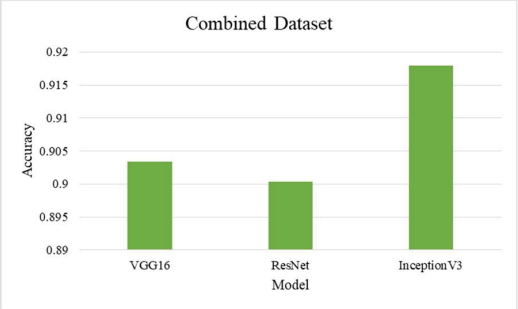

Fig 10. Performance of different transfer learning techniques using the combined dataset

The experimental result illustrates that the combination of different data augmentation techniques based combined dataset gives superior performance than the original dataset and individual augmentation techniques based datasets. Also, the performance of deep learning based dataset is better than the basic manipulation techniques. Finally, the combined dataset was shared in the kaggle open data repository [31].

VII. CONCLUSION

In this research, four different image augmented datasets are proposed to solve the classification problems in plant leaf disease detection. The datasets are Basic image manipulation augmented dataset, DCGAN and WGAN augmented dataset, NST augmented dataset, and a combined augmented dataset. The datasets are used to train the state-of-the-art transfer learning techniques to classify plant leaf diseases. The extensive simulation outcomes gives three important understandings to develop a plant leaf disase detection model. First, the class size balanced augmented dataset gives a better classification accuracy than the original dataset. Second, the performance of deep learning based augmentation technique is higher than the manipulation techniques. At last, the combination of multiple augmentation techniques based dataset gives a superior result than all other datasets. In the future, additional image augmentation techniques such as autoencoders, Random erasing, Feature space augmentation, Adversarial training, and Meta-learning techniques can implement into the plant leaf disease dataset.

REFERENCES

[1] Geetharamani G, Arun Pandian J, Identification of plant leaf diseases using a nine-layer deep convolutional neural network, Comput. Electr. Eng. 76, 323–338, 2019.

[2] Arun Pandian J, Geetharamani G, Mohit Agarwal, Suneet Kumar Gupta, Plant Leaf Disease classification techniques using Leaf Images: A Study, International Journal of ChemTech Research, Vol.12, No.05,pp212-216,2019.

[3] Arun Pandian J, G Geetharamani, identification of Fruit Categories using a nine-layer Deep Convolutional Neural Network, Vol.12, No.05,pp238-246,2019

[4] Alec Radford and Luke Metz and SoumithChintala, Unsupervised Representation Learning with Deep Convolutional Generative Adversarial Networks, arXiv, 2015

[5] Martin Arjovsky and SoumithChintala and Léon Bottou, Wasserstein GAN, arXiv, 2017

[6] Leon A. Gatys and Alexander S. Ecker and Matthias Bethge, A Neural Algorithm of Artistic Style, arXiv, 2015

[7] Tero K, Timo A, Samuli L, Jaakko L. Progressive growing of GANs for improved quality, stability, and variation. CoRR, abs/1710.10196, 2017.

[8] Justin J, Alexandre A, Li FF. Perceptual losses for real-time style transfer and super-resolution. ECCV. 2016;2016:694–711.

[9] In Y, Paul SB, Ekta W. Generative adversarial network in medical imaging: a review. arXiv preprint. 2018.

[10] Maayan F-A, Eyal K, Jacob G, Hayit G. GAN-based data augmentation for improved liver lesion classification. arXiv preprint. 2018.

[11] Laurens M, Geoffrey H. Visualizing data using t-SNE. J Mach Learn Res. 2008;9:2431–56.

[12] Jun-Yan Z, Taesung P, Phillip I, Alexei AE. Unpaired image-to-image translation using cycle-consistent adversarial networks. In: International conference on cmoputer vision (ICCV), 2017

[13] Xinyue Z, Yifan L, Zengchang Q, Jiahong L. Emotion classification with data augmentation using generative adver-sarial networks. CoRR, vol. abs/1711.00648. 2017.

[14] Goodfellow IJ, Erhan D, Carrier PL, Courville A, Mirza M, Hamner B, Cukierski W, Tang Y, Thaler D, Lee DH, et al. Chal-lenges in representation learning: A report on three machine learning contests. In: NIPS. Berlin: Springer; 2013. p.117–24.

[15] Mehdi M, Simon O. Conditional generative adversarial nets. arXiv preprint. 2014

[16] Mario L, Karol K, Marcin M, Olivier B, Sylvain G. Are GANs created equal? A large-scale study. arXiv preprint. 2018.

[17] Swee KL, Yi L, Ngoc-Trung T, Ngai-Man C, Gemma R, Yuval E. DOPING: generative data augmentation for unsuper-vised anomaly detection with GAN. arXiv preprint. 2018.

[18] Alireza M, Jonathon S, Navdeep J, Ian G, Brendan F. Adversarial autoencoders. arXiv preprint. 2015

[19] Tim S, Ian G, Wojciech Z, Vicki C, Alec R, Xi C. Improved techniques for training GANs. arXiv preprint. 2016.

[20] Yanghao L, Naiyan W, Jiaying L, Xiaodi H. Demistifying neural style transfer. arXiv preprint. 2017.

[21] Khizar H. Super-resolution via deep learning. arXiv preprint. 2017

[22] Dmitry U, Andrea V, Victor L. Instance normalization: the missing ingredient for fast stylization. arXiv preprint. 2016.

[23] Philip TJ, Amir AA, Stephen B, Toby B, Boguslaw O. Style augmentation: data augmentation via style randomiza-tion. arXiv e-prints. 2018.

[24] Josh T, Rachel F, Alex R, Jonas S, Wojciech Z, Pieter A. Domain randomization for transferring deep neural networks from simulation to the real world. arXiv preprint. 2017.

[25] Ashish S, Tomas P, Oncel T, Josh S, Wenda W, Russ W. Learning from simulated and unsupervised images through adversarial training. In: Conference on computer vision and pattern recognition, 2017

[26] Stephan RR, Vibhav V, Stefan R, Vladlen K. Playing for data: ground truth from computer games. In: European conference on computer vision (ECCV); 2016

[27] Brostow Gabriel J, Fauqueur Julien, Cipolla Roberto. Semantic object classes in video: a high-definition ground truth database. Pattern RecognLett. 2008;30(2):88–97.

[28] Marius C, Mohamed O, Sebastian R, Timo R, Markus E, Rodrigo B, Uwe F, Stefan R, Bernt S. The cityscape dataset for semantic urban scene understanding. In: CVPR; 2016.

[29] Maayan F-A, Idit D, Eyal K, Michal A, Jacob G, Hayit G. GAN-based synthetic medical image augmentation for increased CNN performance in liver lesion classification. arXiv preprint. 2018.

[30] Changhee H, Hideaki H, Leonardo R, Ryosuke A, Wataru S, Shinichi M, Yujiro F, Giancarlo M, Hideki N. GAN-based synthetic brain mr image generation. In: 2018 IEEE 15th International Symposium on biomedical imaging (ISBI 2018). IEEE, 2011. P. 734

[31] https://www.kaggle.com/arunpandianj/collective-augmented-plant-leaf-disease-dataset

[32] https://www.kaggle.com/arunpandianj/manipulation-augmented-plant-leaf-disease-dataset

[33] https://www.kaggle.com/arunpandianj/gan-augmented-plant-leaf-disease-dataset

[34] J Arun Pandian, Gopal, Geetharamani, "Data for: Identification of Plant Leaf Diseases Using a 9-layer Deep Convolutional Neural Network", Mendeley Data, v1, 2019

[35] https://www.kaggle.com/arunpandianj/nst-augmented-plant-leaf-disease-dataset